

POWERFACTORY

PowerFactory 2024

Technical Reference

ABB Starting

RelFdetabb, TypFdetabb

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

This document describes different configurations and the corresponding dynamic and steady-state behaviour of the *ABB Starting* unit for protection devices in *PowerFactory*.

The main purpose of the *ABB Starting* (*RelFdetabb*) within a *PowerFactory* protection device model is to detect fault conditions and subsequently release the connected distance protection elements. ABB protection devices typically have two functions which can be classified as starting unit, the *General Fault Criteria* and the *Phase Selection*. The former is closer in functionality to typical starting units. The latter is commonly used as an add-on to apply additional constraints for close faults. The *ABB Starting* (*RelFdetabb*) model in *PowerFactory* focuses on the *General Fault Criteria* functionality.

Note: In later device models the naming convention changed to *Phase Selection* and *Phase Preference Logic*.

1.1 Element and Type Configuration

The main setting within the *ABB Starting Type* (*TypFdetabb*) is the *Model* (e:model) selection. It controls the available starting methods and settings, as well as how settings are interpreted. Detailed explanations for each model are given in the following sections.

For each range available in the type, a corresponding setting is available in the element. The element may provide additional settings, such as the phase preference, which depend on the selected model and can not be configured further. In general, all ranges/settings are considered mandatory for the correct function of the model.

2 REL 100

The “REL 100” model does not require additional input. This model is provided for compatibility reasons only. According to [2] the zones only require a minimum current.

This is modelled as overcurrent starting, where the three phase-earth loops are started with the following condition:

$$I_i \geq 0.2 \cdot I_{nom}$$

where:

I_i is the phase current magnitude for phase i (A,B,C) in sec. A (*s:I_i*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

The phase-phase loops are started if both associated phase-earth loops are started.

3 REL 316

The “REL 316” model provides overcurrent and impedance starting [7]. Only one method can be active at the same time.

3.1 Overcurrent Starting

The overcurrent starting determines the start conditions for the individual phases as follows:

$$I_{max} \geq I_{start} \cdot I_{nom}$$
$$I_i \geq 0.8 \cdot I_{max}$$
$$I_i \geq I_{min} \cdot I_{nom}$$

where:

I_i is the phase current magnitude for phase i (A,B,C) in sec. A (*s:I_i*)

I_{max} is the maximum phase current in sec. A

I_{start} is the starting threshold in p.u. (*e:Istart*)

I_{min} is the minimum current threshold in p.u. (*e:Imin*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

In a second step, the phase-wise starting is transformed into loop information. With earth fault indication present (see Section 3.3), the corresponding phase-earth loops are started. Without earth fault indication, the phase-phase loops are started if both associated phases started.

3.2 Impedance Starting

The impedance starting will evaluate either phase-earth or phase-phase impedances, depending on the earth fault indication (see Section 3.3).

Phase-earth loops are evaluated if an earth fault indication is present. The process is separated into two parts. The first part evaluates a minimum current criterion for each phase and evaluates if the impedance $\underline{Z}_i^{'}$ is inside the impedance characteristic (see Figure 3.1).

Minimum current:

$$I_i \geq I_{min} \cdot I_{nom}$$

Initial impedance:

$$\underline{Z}_i^{'} = \frac{\underline{U}_i}{2 \cdot \underline{I}_i}$$

where:

$\underline{Z}_i^{'}$ is the complex impedance for phase i (A,B,C) in sec. Ω

$\underline{U}_i$ is the complex voltage for phase i (A,B,C) in sec. V (*s:Ur_i* $+ j \cdot$ *s:Ui_i*)

$\underline{I}_i$ is the complex current for phase i (A,B,C) in sec. A (*s:Ir_i* $+ j \cdot$ *s:Ii_i*)

I_{min} is the minimum current threshold in p.u. (*e:Imin*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

After determining the temporary starting, the phase preference is applied and the impedance(s) $\underline{Z}_i$ for the resulting loop(s) are again evaluated against the impedance characteristic. Depending on the phase preference, the impedance(s) can be either **phase-earth** or **phase-phase**. The resulting impedances are calculated as follows:

Phase-earth loops:

$$\underline{Z}_i = \frac{\underline{U}_i}{\underline{I}_i + 3\underline{I}_0}$$

Phase-phase loops:

$$\underline{Z}_{ij} = \frac{\underline{U}_i - \underline{U}_j}{\underline{I}_i - \underline{I}_j}$$

where:

$\underline{Z}_i$ is the complex impedance for phase i (A,B,C) in sec. Ω (*c:R:i* $+ j \cdot$ *c:X:i*)

$\underline{Z}_{ij}$ is the complex impedance between phases ij (AB, BC, CA) in sec. Ω (*c:Rl:i* $+ j \cdot$ *c:Xl:i*)

$\underline{U}_i$ is the complex voltage for phase i (A,B,C) in sec. V (*s:Ur_i* $+ j \cdot$ *s:Ui_i*)

$\underline{U}_j$ is the complex voltage for phase j (B,C,A) in sec. V (*s:Ur_j* $+ j \cdot$ *s:Ui_j*)

$\underline{I}_i$ is the complex current for phase i (A,B,C) in sec. A (*s:Ir_i* $+ j \cdot$ *s:Ii_i*)

$\underline{I}_j$ is the complex current for phase j (B,C,A) in sec. A (*s:Ir_j* $+ j \cdot$ *s:Ii_j*)

$3\underline{I}_0$ is the complex neutral current in sec. A (*s:I0x3r* $+ j \cdot$ *s:I0x3i*)

Phase-phase loops are evaluated if no earth fault indication is present. In contrast to the phase-earth loops, only the minimum current criterion for each loop and whether the impedance $\underline{Z}_{ij}$ is inside the impedance characteristic (see Figure 3.1) are evaluated.

Minimum current:

$$I_i \geq I_{min} \cdot I_{nom}$$
$$I_j \geq I_{min} \cdot I_{nom}$$

Impedance:

$$\underline{Z}_{ij} = \frac{\underline{U}_i - \underline{U}_j}{\underline{I}_i - \underline{I}_j}$$

where:

I_i is the phase current magnitude for phase i (A,B,C) in sec. A (*s:I_i*)

I_j is the phase current magnitude for phase j (B,C,A) in sec. A (*s:I_j*)

I_{min} is the minimum current threshold in p.u. (*e:Imin*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

$\underline{Z}_{ij}$ is the complex impedance between phases ij (AB, BC, CA) in sec. Ω (*c:Rl:i* $+ j \cdot$ *c:Xl:i*)

$\underline{U}_i$ is the complex voltage for phase i (A,B,C) in sec. V (*s:Ur_i* $+ j \cdot$ *s:Ui_i*)

$\underline{U}_j$ is the complex voltage for phase j (B,C,A) in sec. V (*s:Ur_j* $+ j \cdot$ *s:Ui_j*)

$\underline{I}_i$ is the complex current for phase i (A,B,C) in sec. A (*s:Ir_i* $+ j \cdot$ *s:Ii_i*)

$\underline{I}_j$ is the complex current for phase j (B,C,A) in sec. A (*s:Ir_j* $+ j \cdot$ *s:Ii_j*)

3.3 Phase Selection

The “Earth fault detection” controls whether phase-phase or phase-earth loops are started. Depending on the selected mode (*e:GndFmode*), either the $3I_0$ and/or $3U_0$ must satisfy the following conditions to indicate an earth fault:

I0:

$$3I_0 \geq 3I_{0,min} \cdot I_{nom}$$
$$3I_0 \geq 0.25 \cdot I_{max}$$

U0:

$$3U_0 \geq 3U_{0,min} \cdot U_{nom}$$

where:

$3I_0$ is the magnitude of the neutral current in sec. A (*s:I0x3*)

$3I_{0,min}$ is the zero sequence current threshold in p.u. (*e:I0x3min*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

I_{max} is the maximum phase current in sec. A

$3U_0$ is the magnitude of the residual voltage in sec. V (*s:U0x3*)

$3u_{0,min}$ is the zero sequence voltage threshold in p.u. (*e:U0x3min*)

U_{nom} is the nominal secondary voltage in sec. V (*e:Unom* of the connected measurement)

The “Cross-country fault detection” defines the phase preference for 2-phase-to-ground faults in compensated or isolated networks. Please refer to [7] for more information.

3.4 Impedance Characteristic

The impedance characteristic used for phase-phase and phase-earth measurement loops is shown in Figure 3.1.

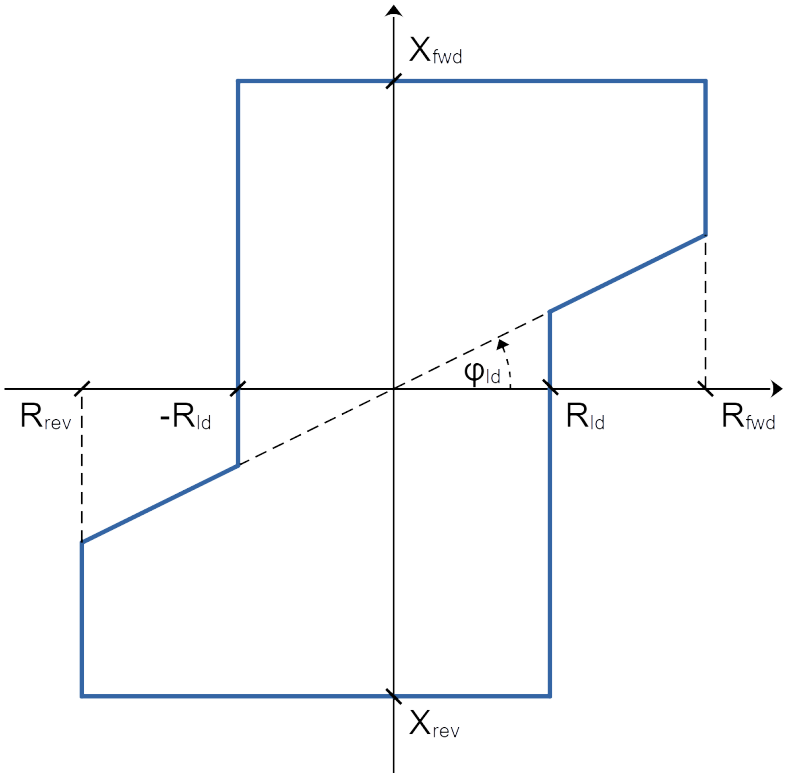


Figure 3.1: Impedance starting characteristic for REL 316

Note: In contrast to other impedance characteristics described in this document, the “REL 316” model expects the reverse impedances as negative values.

The points indicated in Figure 3.1 are defined as follows:

X_{fwd} Reactive reach in forward direction (*e:X1FwPP*)

X_{rev} Reactive reach in reverse direction (*e:X1RvPP*)

R_{fwd} Resistive reach in forward direction (*e:RFPP*)

R_{rev} Resistive reach in reverse direction (*e:RRvPP*)

R_{ld} Load resistance (*e:RLd*)

φ_{ld} Load angle (*e:ARGLd*)

4 REL 511

The “REL 511” model provides overcurrent and impedance starting, which can be used independently [4], [5].

4.1 Overcurrent Starting

The overcurrent starting determines the starting conditions for the individual phases and the neutral starting condition as follows:

$$I_i \geq \frac{I_P}{100} \cdot I_{nom}$$
$$3I_0 \geq \frac{I_N}{100} \cdot I_{nom}$$

where:

I_i is the phase current magnitude for phase i (A,B,C) in sec. A (*s:I_i*)

$3I_0$ is the neutral current magnitude in sec. A (*s:I0x3*)

I_P is the phase current threshold in % (*e:IPM*)

I_N is the neutral current threshold in % (*e:INM*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

For a **phase-earth** loop to start, the condition for the phase current threshold and the neutral current threshold must be satisfied. For a **phase-phase** loop to start, both phase current criteria must be satisfied. In addition, the corresponding measurement system must be released by the phase selection (see Section 4.3).

4.2 Impedance Starting

The impedance starting will evaluate phase-earth and/or phase-phase impedances, depending on the phase selection (see Section 4.3).

Phase-earth loops are calculated with two separate compensation factors, one for forward and one for reverse direction, and checked against the impedance characteristic (see Figure 4.1). For a loop to start, both impedances must be inside the characteristic, the minimum current criterion must be satisfied and the phase-earth measurement must be released by the phase selection (see Section 4.3). The impedances and the minimum current criterion are calculated as follows:

Minimum current:

$$I_i \geq 0.5 \cdot \frac{I_{min}}{100} \cdot I_{nom}$$

Impedances:

$$\underline{Z}_{i,fwd} = \frac{\underline{U}_i}{\underline{I}_i \cdot (1 + K_{NF})}$$
$$\underline{Z}_{i,rev} = \frac{\underline{U}_i}{\underline{I}_i \cdot (1 + K_{NR})}$$

where:

I_i is the phase current magnitude for phase i (A,B,C) in sec. A (*s:I_i*)

I_{min} is the minimum current threshold in % (*e:IMinOp*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

$\underline{U}_i$ is the complex voltage for phase i (A,B,C) in sec. V (*s:Ur_i* $+ j \cdot$ *s:Ui_i*)

$\underline{I}_i$ is the complex current for phase i (A,B,C) in sec. A (*s:Ir_i* $+ j \cdot$ *s:Ii_i*)

K_{NF} is the compensation factor in forward direction (see Section 4.4)

K_{NF} is the compensation factor in reverse direction (see Section 4.4)

Note: The calculation quantities *c:R:i* and *c:X:i* show the resistive and reactive parts of $\underline{Z}_{i,fwd}$ if the resistive part is greater or equal to zero; otherwise the resistive and reactive parts of $\underline{Z}_{i,rev}$ are shown.

Phase-phase loops only use one calculated impedance. For a loop to start, the impedance must be inside the impedance characteristic (see Figure 4.1), the minimum current criterion must be satisfied and the phase-phase measurement must be released by the phase selection (see Section 4.3). The impedance and the minimum current criterion are calculated as follows:

Minimum current:

$$|\underline{I}_i - \underline{I}_j| \geq \frac{I_{min}}{100} \cdot I_{nom}$$

Impedance:

$$\underline{Z}_{ij} = \frac{\underline{U}_i - \underline{U}_j}{\underline{I}_i - \underline{I}_j}$$

where:

$\underline{I}_i$ is the complex current for phase i (A,B,C) in sec. A (*s:Ir_i* $+ j \cdot$ *s:Ii_i*)

$\underline{I}_j$ is the complex current for phase j (B,C,A) in sec. A (*s:Ir_j* $+ j \cdot$ *s:Ii_j*)

I_{min} is the minimum operating current threshold in p.u. (*e:IminOp*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

$\underline{Z}_{ij}$ is the complex impedance between phases ij (AB, BC, CA) in sec. Ω (*c:Rl:i* $+ j \cdot$ *c:Xl:i*)

$\underline{U}_i$ is the complex voltage for phase i (A,B,C) in sec. V (*s:Ur_i* $+ j \cdot$ *s:Ui_i*)

$\underline{U}_j$ is the complex voltage for phase j (B,C,A) in sec. V (*s:Ur_j* $+ j \cdot$ *s:Ui_j*)

4.3 Phase Selection

The phase selection releases the phase-earth and phase-phase measurement systems as follows:

Phase-earth:

$$3I_0 \geq 0.5 \cdot \frac{I_{min}}{100} \cdot I_{nom} \qquad \text{AND} \qquad 3I_0 \geq \frac{I_{PE}}{100} \cdot I_{max}$$

Phase-phase:

$$3I_0 < 0.2 \cdot I_{nom} \qquad \text{OR} \qquad 3I_0 < \frac{I_{PP}}{100} \cdot I_{max}$$

where:

$3I_0$ is the magnitude of the neutral current in sec. A (*s:I0x3*)

I_{min} is the minimum operating current threshold in %. (*e:IminOp*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

I_{max} is the maximum phase current in sec. A

I_{PE} is the minimum current for phase-earth release in % (*e:INReleasePE*)

I_{PP} is the minimum current for phase-phase blocking in % (*e:INBlockPP*)

The “Cross-country fault detection” determines the presence of 2-phase-to-ground faults, also called double earth faults, and applies the phase preference as configured. A 2-phase-to-ground fault is present if either of the following conditions is satisfied:

- $3I_0 \geq \frac{I_N}{100} \cdot I_{nom}$ for more than 150 ms;
- $3I_0 \geq \frac{I_N}{100} \cdot I_{nom}$ and $3U_0 \geq \frac{U_N}{100} \cdot \frac{U_{nom}}{\sqrt{3}}$ for more than 100ms;
- $3I_0 \geq \frac{I_N}{100} \cdot I_{nom}$ and either any phase-phase voltage or any two phase-earth voltages fall below the respective thresholds.

where:

$3I_0$ is the magnitude of the neutral current in sec. A (*s:I0x3*)

I_N is the zero sequence current threshold in % (*e:INmM*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

$3U_0$ is the magnitude of the residual voltage in sec. B (*s:U0x3*)

U_N is residual voltage threshold in % (*e:U0x3M*)

U_{nom} is the nominal secondary voltage in sec. V (*e:Unom* of the connected measurement)

Note: For steady-state calculations, such as load flow or short-circuit, only the zero sequence current criterion is evaluated, but without the time component.

The phase preference is applied as described in [4], [5]. The phase preference only applies to the loop signal (*s:yloop*), not to the phase-wise starting signals.

The “Time delayed trip” generates a dedicated tripping signal directly from the starting unit. The two separate timers for phase-earth (*e:tPE*) and (*e:tPP*) are started if either impedance or overcurrent starting detect a fault for a corresponding loop. The tripping output (*s:yout*) corresponds to whichever timer expires first.

4.4 Impedance Characteristic

The impedance characteristic used for phase-phase and phase-earth measurement loops is shown in Figure 4.1. While the general shape is identical, different parameters are used to determine the characteristic.

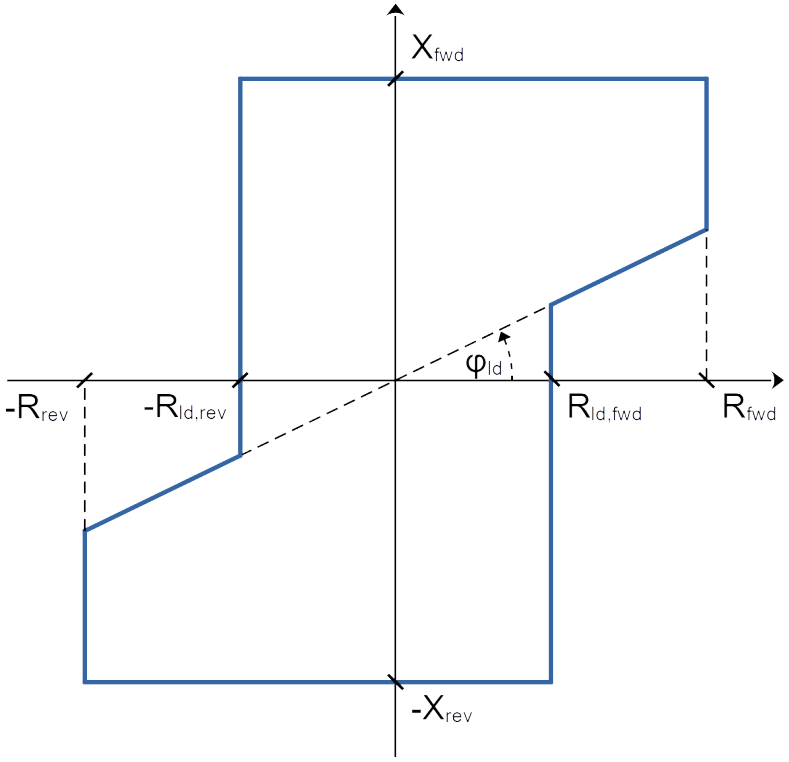


Figure 4.1: Impedance starting characteristic for REL 511

The points indicated in Figure 4.1 are calculated as follows for the **phase-phase** characteristic:

X_{fwd} Reactive reach in forward direction (*e:X1FwPP*)

X_{rev} Reactive reach in reverse direction (*e:X1RvPP*)

R_{fwd}**,** R_{rev} Resistive reach in forward and reverse direction

$$R_{fwd} = -R_{rev} = \frac{R_{PP}}{2}$$

where R_{PP} is the resistive reach for phase-phase loops (*e:RFPP*)

$R_{ld,fwd}$**,** $R_{ld,rev}$ Load resistance in forward and reverse direction

$$R_{ld,fwd} = -R_{ld,rev} = \frac{R_L}{2}$$

where R_L is the Load resistance (*e:RLd*)

φ_{ld} Load angle (*e:ARGLd*)

The points indicated in Figure 4.1 are calculated as follows for the **phase-earth** characteristic:

X_{fwd} Reactive reach in forward direction (*e:X1FwPE*)

X_{rev} Reactive reach in reverse direction (*e:X1RvPE*)

R_{fwd} Resistive reach in forward direction

$$R_{fwd} = \frac{R_{PE}}{1 + K_{NF}}$$

where R_{PE} is the resistive reach for phase-earth loops (*e:RFPE*)

R_{rev} Resistive reach in reverse direction

$$R_{rev} = -\frac{R_{PE}}{1 + K_{NR}}$$

where R_{PE} is the resistive reach for phase-earth loops (*e:RFPE*)

$R_{ld,fwd}$ Load resistance in forward direction

$$R_{ld,fwd} = \frac{R_L}{1 + K_{NF}}$$

where R_L is the load resistance (*e:RLd*)

$R_{ld,rev}$ Load resistance in reverse direction

$$R_{ld,rev} = -\frac{R_L}{1 + K_{NR}}$$

where R_L is the load resistance (*e:RLd*)

φ_{ld} Load angle (*e:ARGLd*)

The compensation factors K_{NF} and K_{NR} are calculated as follows:

$$K_{NF} = \frac{X_{0,fwd} - X_{1,fwd}}{3 \cdot X_{1,fwd}}$$
$$K_{NR} = \frac{X_{0,rev} - X_{1,rev}}{3 \cdot X_{1,rev}}$$

where:

$X_{0,fwd}$ is the zero sequence reactance in forward direction (*e:X0FwPE*)

$X_{1,fwd}$ is the positive sequence reactance in forward direction (*e:X1FwPE*)

$X_{0,rev}$ is the zero sequence reactance in reverse direction (*e:X0RvPE*)

$X_{1,rev}$ is the positive sequence reactance in reverse direction (*e:X1RvPE*)

5 REL 670

The “REL 670” model provides overcurrent and impedance starting, which can be used independently [3], [6].

5.1 Overcurrent Starting

The overcurrent starting determines the starting conditions for the individual phases and the neutral starting condition as follows:

$$I_i \geq \frac{I_P}{100} \cdot I_{nom}$$
$$3I_0 \geq \frac{I_N}{100} \cdot I_{nom}$$

where:

I_i is the phase current magnitude for phase i (A,B,C) in sec. A (*s:I_i*)

$3I_0$ is the neutral current magnitude in sec. A (*s:I0x3*)

I_P is the phase current threshold in % (*e:IPM*)

I_N is the neutral current threshold in % (*e:INM*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

For a **phase-earth** loop to start, the condition for the phase current threshold and the neutral current threshold must be satisfied. For a **phase-phase** loop to start, both phase current criteria must be satisfied. In addition, the corresponding measurement system must be released by the phase selection (see Section 5.3).

5.2 Impedance Starting

The impedance starting evaluates phase-earth and/or phase-phase impedances, depending on the phase selection (see Section 5.3).

Phase-earth loops are calculated as described below and checked against the impedance characteristic (see Figure 5.1). For a loop to start, the impedance must be inside the characteristic, the minimum current criterion must be satisfied and the phase-earth measurement must be released by the phase selection (see Section 5.3). The impedances and the minimum current criterion are calculated as follows:

Minimum current:

$$I_i \geq \frac{I_{min}}{100} \cdot I_{nom}$$

Impedance:

$$\underline{Z}_i = \frac{\underline{U}_i}{\underline{I}_i}$$

where:

I_i is the phase current magnitude for phase i (A,B,C) in sec. A (*s:I_i*)

I_{min} is the phase-earth operating current threshold in % (*e:IMinOpPE*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

$\underline{Z}_i$ is the complex impedance for phase i (A,B,C) in sec. Ω (*c:R:i* $+ j \cdot$ *c:X:i*)

$\underline{U}_i$ is the complex voltage for phase i (A,B,C) in sec. V (*s:Ur_i* $+ j \cdot$ *s:Ui_i*)

$\underline{I}_i$ is the complex current for phase i (A,B,C) in sec. A (*s:Ir_i* $+ j \cdot$ *s:Ii_i*)

Phase-phase loops are calculated as described below and checked against the impedance characteristic (see Figure 5.1). For a loop to start, the impedance must be inside the characteristic, the minimum current criterion must be satisfied and the phase-phase measurement must be released by the phase selection (see Section 5.3). The impedances and the minimum current criterion are calculated as follows:

Minimum current:

$$I_j \geq 0.5 \cdot \frac{I_{min}}{100} \cdot I_{nom}$$

Impedance:

$$\underline{Z}_{ij} = \frac{\underline{U}_i - \underline{U}_j}{-2 \cdot \underline{I}_j}$$

where:

I_j is the phase current magnitude for phase j (B,C,A) in sec. A (*s:I_j*)

I_{min} is the phase-phase operating current threshold in % (*e:IMinOpPP*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

$\underline{Z}_{ij}$ is the complex impedance between phases ij (AB, BC, CA) in sec. Ω (*c:RI:i* $+ j \cdot$ *c:XI:i*)

$\underline{I}_j$ is the complex current for phase j (B,C,A) in sec. A (*s:Ir_j* $+ j \cdot$ *s:Ii_j*)

$\underline{U}_i$ is the complex voltage for phase i (A,B,C) in sec. V (*s:Ur_i* $+ j \cdot$ *s:Ui_i*)

$\underline{U}_j$ is the complex voltage for phase j (B,C,A) in sec. V (*s:Ur_j* $+ j \cdot$ *s:Ui_j*)

Note: Due to the way the phase-phase impedances are calculated the effective impedance characteristic for 3-phase faults is larger (by a factor of $2/\sqrt{3}$) and rotated counter-clockwise by $30°$ [6].

5.3 Phase Selection

The phase selection releases the phase-earth and phase-phase measurement systems as follows:

Phase-earth:

$$3I_0 \geq 0.5 \cdot \frac{I_{min}}{100} \cdot I_{nom} \qquad \text{AND} \qquad 3I_0 \geq \frac{I_{PE}}{100} \cdot I_{max}$$

Phase-phase:

$$3I_0 < \frac{I_{min}}{100} \cdot I_{nom} \qquad \text{OR} \qquad 3I_0 < \frac{I_{PP}}{100} \cdot I_{max}$$

where:

$3I_0$ is the magnitude of the neutral current in sec. A (*s:I0x3*)

I_{min} is the phase-earth operating current threshold in % (*e:IminOpPE*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

I_{max} is the maximum phase current in sec. A

I_{PE} is the minimum current for phase-earth release in % (*e:INReleasePE*)

I_{PP} is the minimum current for phase-phase blocking in % (*e:INBlockPP*)

The “Cross-country fault detection” determines the presence of 2-phase-to-ground faults, also called double earth faults, and applies the phase preference as configured. A 2-phase-to-ground fault is present if either of the following conditions is satisfied:

- $3I_0 \geq \frac{I_N}{100} \cdot I_{nom}$ for more than 100 ms;
- $3I_0 \geq \frac{I_N}{100} \cdot I_{nom}$ and $3U_0 \geq \frac{U_N}{100} \cdot \frac{U_{nom}}{\sqrt{3}}$ for more than 100ms;
- $3I_0 \geq \frac{I_N}{100} \cdot I_{nom}$ and either any phase-phase voltage or any two phase-earth voltages fall below the respective thresholds.

where:

$3I_0$ is the magnitude of the neutral current in sec. A (*s:I0x3*)

I_N is the zero sequence current threshold in % (*e:INmM*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

$3U_0$ is the magnitude of the residual voltage in sec. B (*s:U0x3*)

U_N is residual voltage threshold in % (*e:U0x3M*)

U_{nom} is the nominal secondary voltage in sec. V (*e:Unom* of the connected measurement)

Note: For steady-state calculations, such as load flow or short-circuit, only the zero sequence current criterion is evaluated, but without the time component.

The phase preference is applied as described in [3], [6]. The phase preference only applies to the loop signal (*s:yloop*), not to the phase-wise starting signals.

The “Time delayed trip” generates a dedicated tripping signal directly from the starting unit. The two separate timers for phase-earth (*e:tPE*) and (*e:tPP*) are started if either impedance or overcurrent starting detect a fault for a corresponding loop. The tripping output (*s:yout*) corresponds to whichever timer expires first.

5.4 Impedance Characteristic

The impedance characteristic used for phase-phase and phase-earth measurement loops is shown in Figure 5.1. While the general shape is identical, different parameters are used to determine the characteristic.

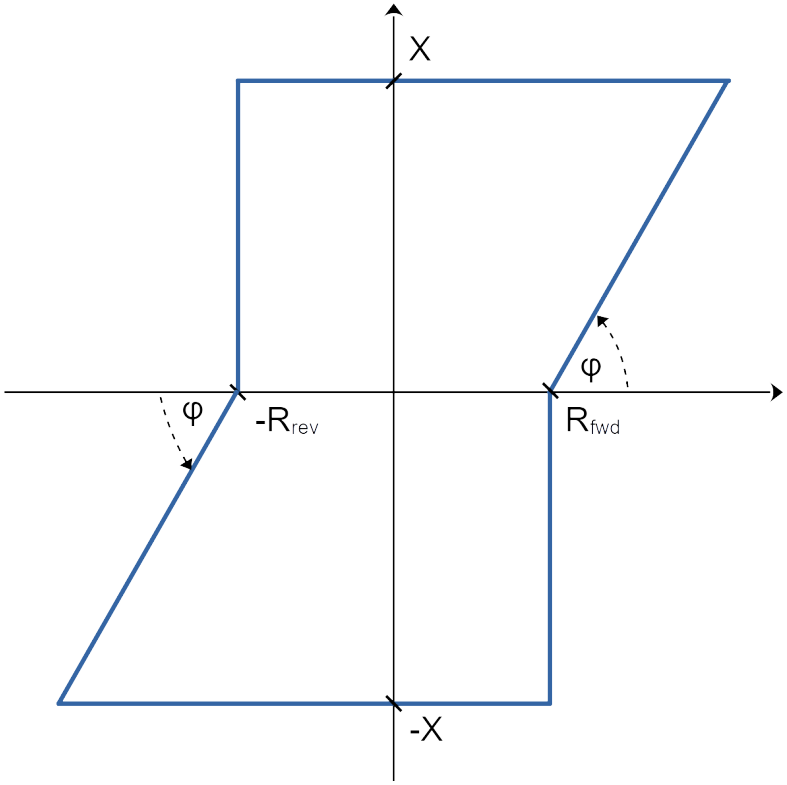


Figure 5.1: Impedance starting characteristic for REL 670

The points indicated in Figure 5.1 are calculated as follows for the **phase-phase** characteristic:

X Reactive reach in forward and reverse direction (*e:X1PP*)

R_{fwd} Resistive reach in forward direction

$$R_{fwd} = \frac{R_{PP}}{2}$$

where R_{PP} is the resistive reach for phase-phase loops (*e:RFPP*)

R_{rev} Resistive reach in reverse direction

$$R_{rev} = \frac{R_{PP,rev}}{2}$$

where $R_{PP,rev}$ is the reverse resistive reach for phase-phase loops (*e:RFRvPP*)

φ Fixed zone inclination angle $\varphi = 60°$

The points indicated in Figure 5.1 are calculated as follows for the **phase-earth** characteristic:

X Reactive reach in forward and reverse direction

$$X = \frac{2 \cdot X_{1,PP} + X_{0,PE}}{3}$$

where:

$X_{1,PP}$ is the reactive reach for phase-phase loops (*e:X1PP*)

$X_{0,PE}$ is the zero sequence reactance for phase-earth loops (*e:X0PE*)

R_{fwd} Resistive reach in forward direction for phase-earth loops (*e:RFPE*)

R_{rev} Resistive reach in reverse direction for phase-earth loops (*e:RFRvPE*)

φ Fixed zone inclination angle $\varphi = 60°$

Note: The phase-earth characteristic of the “REL 670” is shown in the per-loop domain by default. To facilitate comparison with devices from other manufacturers in R-X plots, a dedicated option “Force Positive Sequence Representation” can be enabled. Please see the *PowerFactory* User Manual for more information.

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6 REX 640

The “REX 640” model provides overcurrent, voltage dependent overcurrent, impedance or a combination of overcurrent and impedance starting [1]. Only one method can be active at the same time.

6.1 Overcurrent Starting

The overcurrent starting determines the start conditions for the individual phases as follows:

$$I_i \geq I_{start} \cdot I_{nom}$$

where:

I_i is the phase current magnitude for phase i (A,B,C) in sec. A (*s:I_i*)

I_{start} is the starting threshold in p.u. (*e:Istart*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

In a second step, the phase-wise starting is transformed into loop information. With earth fault indication present (see Section 6.4), the corresponding phase-earth loops are started. The phase-phase loops are started if both associated phases started.

6.2 Voltage Dependent Overcurrent Starting

In principle the voltage dependent overcurrent starting works like the overcurrent starting, but with an additional sensitive stage which is activated if the voltage drops below a threshold.

Phase-earth loops:

$$I_i \geq I_{start} \cdot I_{nom}$$
$$I_i \geq I_{start,U<} \cdot I_{nom} \quad \text{AND} \quad U_i \leq U_{PE} \cdot U_{nom}$$

Phase-phase loops:

$$I_i \geq I_{start} \cdot I_{nom} \quad \text{AND} \quad I_j \geq I_{start} \cdot I_{nom}$$
$$I_i \geq I_{start,U<} \cdot I_{nom} \quad \text{AND} \quad I_j \geq I_{start,U<} \cdot I_{nom} \quad \text{AND} \quad U_{ij} \leq U_{PP} \cdot U_{nom}$$

where:

I_i is the phase current magnitude for phase i (A,B,C) in sec. A (*s:I_i*)

I_j is the phase current magnitude for phase j (B,C,A) in sec. A (*s:I_j*)

I_{start} is the starting threshold in p.u. (*e:Istart*)

$I_{start,U<}$ is the voltage dependent starting threshold in p.u. (*e:IstartVDep*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

U_i is the phase voltage magnitude for phase i (A,B,C) in sec. V (*s:I_i*)

U_{ij} is the phase-phase voltage magnitude between phase ij (AB,BC,CA) in sec. V

$$U_{ij} = |\underline{U}_i - \underline{U}_j|$$

where:

$\underline{U}_i$ is the complex voltage for phase i (A,B,C) in sec. V (*s:Ur_i* $+ j \cdot$ *s:Ui_i*)

$\underline{U}_j$ is the complex voltage for phase j (B,C,A) in sec. V (*s:Ur_j* $+ j \cdot$ *s:Ui_j*)

U_{PE} is the phase-earth voltage threshold in p.u. (*e:UPEVDep*)

U_{PP} is the phase-phase voltage threshold in p.u. (*e:UPPVDep*)

U_{nom} is the nominal secondary voltage in sec. V (*e:Unom* of the connected measurement)

6.3 Impedance Starting

The impedance starting evaluates phase-earth and/or phase-phase impedances, depending on the phase selection (see Section 6.4).

Phase-earth and phase-phase loops are calculated as described below and checked against the configured impedance characteristic (see Figure 6.1 or Figure 6.2). For a loop to start, the impedance must be inside the characteristic and the corresponding measurement must be released by the phase selection (see Section 6.4). The impedances are calculated as follows:

$$\underline{Z}_i = \frac{\underline{U}_i}{\underline{I}_i}$$

$$\underline{Z}_{ij} = \frac{\underline{U}_i - \underline{U}_j}{\underline{I}_i - \underline{I}_j}$$

where:

$\underline{Z}_i$ is the complex impedance for phase i (A,B,C) in sec. Ω (*c:R:i* $+ j \cdot$ *c:X:i*)

$\underline{Z}_{ij}$ is the complex impedance between phase ij (AB,BC,CA) in sec. Ω (*c:RI:i* $+ j \cdot$ *c:XI:i*)

$\underline{U}_i$ is the complex voltage for phase i (A,B,C) in sec. V (*s:Ur_i* $+ j \cdot$ *s:Ui_i*)

$\underline{U}_j$ is the complex voltage for phase j (B,C,A) in sec. V (*s:Ur_j* $+ j \cdot$ *s:Ui_j*)

$\underline{I}_i$ is the complex current for phase i (A,B,C) in sec. A (*s:Ir_i* $+ j \cdot$ *s:Ii_i*)

$\underline{I}_j$ is the complex current for phase j (B,C,A) in sec. A (*s:Ir_j* $+ j \cdot$ *s:Ii_j*)

6.4 Phase Selection

The “Earth fault detection” controls whether phase-phase or phase-earth loops are started. Depending on the selected mode (*e:GndFmode*), either the $3I_0$ and/or $3U_0$ must satisfy the following conditions to indicate an earth fault:

I0:

$$3I_0 \geq 3I_{0,min} \cdot I_{nom}$$
$$3I_0 \geq 0.25 \cdot I_{max}$$

U0:

$$3U_0 \geq 3U_{0,min} \cdot U_{nom}$$

where:

$3I_0$ is the magnitude of the neutral current in sec. A (*s:I0x3*)

$3I_{0,min}$ is the zero sequence current threshold in p.u. (*e:I0x3min*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

I_{max} is the maximum phase current in sec. A

$3U_0$ is the magnitude of the residual voltage in sec. V (*s:U0x3*)

$3u_{0,min}$ is the zero sequence voltage threshold in p.u. (*e:U0x3min*)

U_{nom} is the nominal secondary voltage in sec. V (*e:Unom* of the connected measurement)

The “Cross-country fault detection” determines the presence of 2-phase-to-ground faults, also called double earth faults, and applies the phase preference as configured. Depending on the configured system grounding (*e:isysstar*) the following applies:

With “High impedance” grounding, a 2-phase-to-ground fault is present if an earth fault indication is present and either of the following conditions is satisfied:

- $3I_0 \geq I_{NXC} \cdot I_{nom}$ for more than 100 ms;
- $U_{ij} \leq U_{PP,XC} \cdot U_{nom}$

where:

$3I_0$ is the magnitude of the neutral current in sec. A (*s:I0x3*)

I_{NXC} is the residual current threshold for cross-country faults in p.u. (*e:INXC*)

I_{nom} is the nominal secondary current in sec. A (*e:Inom* of the connected measurement)

U_{ij} is the phase-phase voltage magnitude between phase ij (AB,BC,CA) in sec. V

$$U_{ij} = |\underline{U}_i - \underline{U}_j|$$

where:

$\underline{U}_i$ is the complex voltage for phase i (A,B,C) in sec. V (*s:Ur_i* $+ j \cdot$ *s:Ui_i*)

$\underline{U}_j$ is the complex voltage for phase j (B,C,A) in sec. V (*s:Ur_j* $+ j \cdot$ *s:Ui_j*)

$U_{PP,XC}$ is the phase-phase voltage threshold for cross-country faults in p.u. (*e:UPPXC*)

U_{nom} is the nominal secondary voltage in sec. V (*e:Unom* of the connected measurement)

The phase preference is applied as described in [3], [6]. The phase preference only applies to the loop signal (*s:yloop*), not to the phase-wise starting signals.

With “Low impedance” grounding, the configured phase preference (*e:iPhPhE*) is applied to the loop signal (*s:yloop*) if two phase-to-ground loops are started. If only one or all three phase-to-ground loops are started, no action is taken.

The “Time delayed trip” generates a dedicated tripping signal directly from the starting unit. The timer is started if at least one starting loop is detected.

6.5 Impedance Characteristic

The impedance characteristic used for phase-phase and phase-earth measurement loops can be either rectangular (see Figure 6.1) or circular (see Figure 6.2)). While the general shape is identical, different parameters are used to determine the characteristic for phase-phase and phase-earth characteristics.

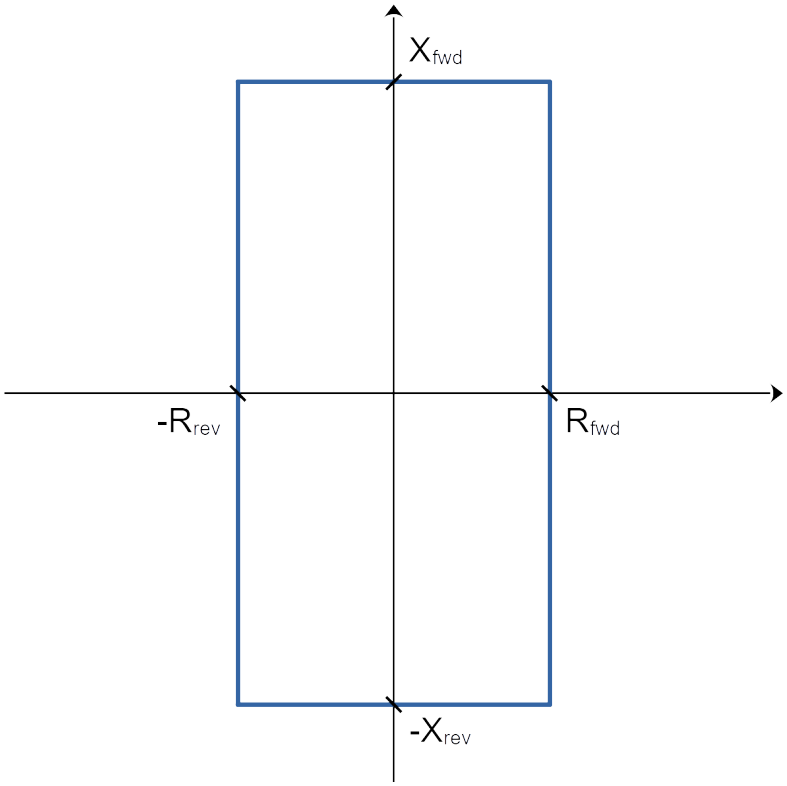


Figure 6.1: Rectangular impedance starting characteristic for REX 640

The points indicated in Figure 6.1 are calculated as follows for the **phase-phase** characteristic:

X_{fwd} Reactive reach in forward direction for phase-phase loops (*e:X1FwPP*)

X_{rev} Reactive reach in reverse direction for phase-phase loops (*e:X1RvPP*)

R_{fwd} Resistive reach in forward direction for phase-phase loops (*e:RFPP*)

R_{rev} Resistive reach in reverse direction for phase-phase loops (*e:RFRvPP*)

The points indicated in Figure 6.1 are calculated as follows for the **phase-earth** characteristic:

X_{fwd} Reactive reach in forward direction for phase-earth loops (*e:X1FwPE*)

X_{rev} Reactive reach in reverse direction for phase-earth loops (*e:X1RvPE*)

R_{fwd} Resistive reach in forward direction for phase-earth loops (*e:RFPE*)

R_{rev} Resistive reach in reverse direction for phase-earth loops (*e:RFRvPE*)

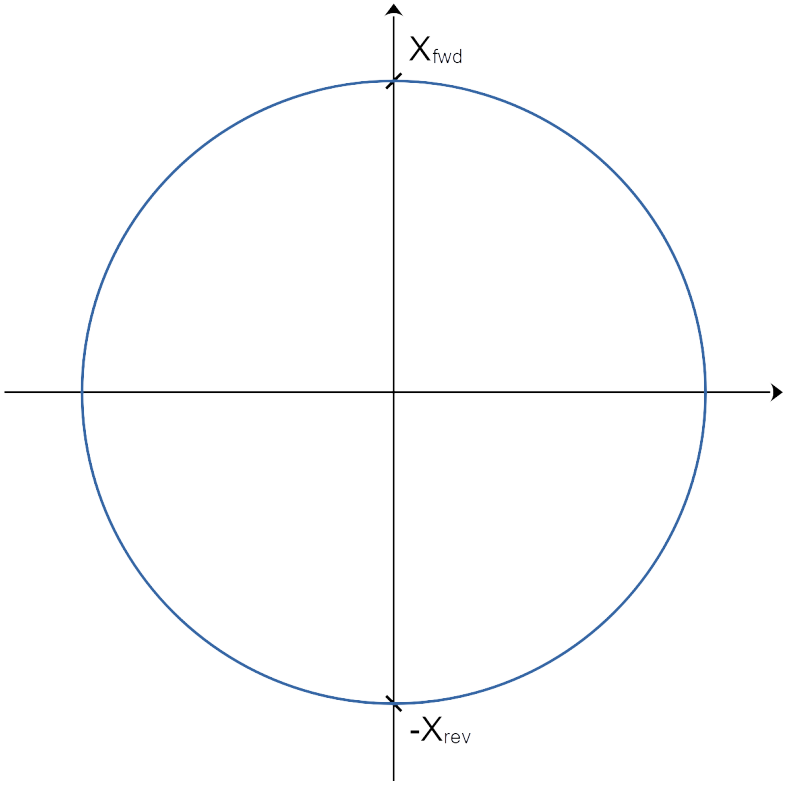


Figure 6.2: Circular impedance starting characteristic for REX 640

The points indicated in Figure 6.2 are calculated as follows for the **phase-phase** characteristic:

X_{fwd} Reactive reach in forward direction for phase-phase loops (*e:X1FwPP*)

X_{rev} Reactive reach in reverse direction for phase-phase loops (*e:X1RvPP*)

The points indicated in Figure 6.2 are calculated as follows for the **phase-earth** characteristic:

X_{fwd} Reactive reach in forward direction for phase-earth loops (*e:X1FwPE*)

X_{rev} Reactive reach in reverse direction for phase-earth loops (*e:X1RvPE*)

7 Calculation Models

This section provides a brief explanation for the calculation models, specifically the interpretation of some of the output signals.

The starting signals (*s:ystart*, *s:ys_A*, *s:ys_B*, *s:ys_C* and *s:ysall*) indicate fault detection on any phase, a particular phase or all three phases respectively. The overcurrent (*s:yistart*) and impedance starting (*s:yzstart*) indicate the fault detection by the overcurrent or impedance starting respectively. For “REX 640” the overcurrent starting signal also covers the voltage dependent overcurrent starting.

The fault loop signal (*s:yloop*) provides the detailed fault loop information.

Bit	0	1	2	3	4	5
Loop	A	B	C	AB	BC	CA

Table 7.1: Bit-wise layout of the loop signal

The numerical value of the loop signal can then be calculated as:

$$loop = \sum_{n=0}^{5} (bit(n) \cdot 2^n)$$

For example, a combination of the A, B and AB loops can be expressed as `001011` (binary is written from right to left) and therefore results in a loop value of:

$$1 \cdot 2^0 + 1 \cdot 2^1 + 0 \cdot 2^2 + 1 \cdot 2^3 + 0 \cdot 2^4 + 0 \cdot 2^5 = 11$$

Note: This calculation can also be performed by the Microsoft Windows calculator application when it is set to Programmer.

Notable values of the loop signal:

0**:** *None*

7**:** *A, B, C*

56**:** *AB, BC, CA*

63**:** *A, B, C, AB, BC, CA*

The tripping signal (*s:yout*) indicates the expiration of the tripping timer. This signal is unused for “REL 316”.

8 References

[1] ABB Distribution Solutions. *RELION PROTECTION AND CONTROL REX640 Technical Manual Revision E*, December 2020. 1MRS759142.

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[3] ABB Power Grids Sweden AB. *Relion 670 SERIES Line distance protection REL670 Version 2.2 IEC Application manual Revision K*, June 2021. 1MRK 506 369-UEN.

[4] ABB Power Technologies AB. *Application manual ProtectIT Line distance protection terminal REL 511*2.5 Revision B*, December 2006. 1MRK 506 168-UEN.

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[7] ABB Switzerland Ltd. *RE.316*4 Numerical Protection and Control Devices Operating Instructions*, October 2004. 1KHA000835-UEN.

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